Intro to TypeScript

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Course repository: https://github.com/moonhighway/intro-to-typescript

Required: Node.js >= v18.0.0

Recommended: Visual Studio Code (with GitHub Copilot disabled)

Setup

If you only have a README in your cloned copy of the repository, run:

```
git pull
```

Install dependencies by running:

```
npm run setup
```

Agenda

- **Three sections:** A World of Types, Types in Action, Flexible Types
- **We'll build an app:** Trip Planner command-line interface (CLI)

Goals

- Learn what TypeScript is and how to use it
- Configure a TypeScript project
- Get comfortable with the fundamentals
- Tackle common "gotchas" when using TypeScript

A World of Types

What is TypeScript?

- "TypeScript is **JavaScript with syntax for types"**

```
let myFavoriteFood = 'Pozole Rojo';
```

```
let myFavoriteFood: string = 'Pozole Rojo';
```

- Strongly typed programming language
- Released publicly by Microsoft in 2012
- TypeScript = JavaScript + types

Basic Types

- Types for JavaScript primitives: `string`, `number`, `boolean`, `null`
- `any`

Objects

- Interface
- Type alias
- Optional properties: `?`
- `readonly` properties

Arrays

- Two types of syntax
- `Array<type>`
- `type[]`
- Tuples

Benefits of TypeScript

- **“The main benefit of TypeScript is that it can highlight unexpected behavior in your code, lowering the chance of bugs.**”

  — *TypeScript docs*

- Types as a form of documentation
- Safer code refactoring
- Can use types as a contract between different parts of our application

Type Declarations

1. Add types into project code that we write

2. Built-in to third-party package

3. Provided by an `@types` packages (DefinitelyTyped project)

How do we run TypeScript code?

- We don't, we compile it!
- TypeScript code compiles to JavaScript
- Types are stripped away at compile time
- Then we run the compiled JavaScript

TypeScript Project Configuration

- TSConfig
  - TSConfig reference
  - TSConfig bases
- Linting
  - typescript-eslint

Compilation & Execution

- Tools that can compile TypeScript:

  TypeScript compiler (`tsc`), Vite, esbuild, swc

- Tools that compile TypeScript **and** run the code:

  Deno, Bun, tsx, tsm, ts-node

Types in Action

Functions

- Parameter types
- Return types
- `async` functions and promises

Classes

Union types

- Pipe operator: |
- Anywhere you can put a type, you can put a union type

Type Narrowing

- `typeof` type guard
- `instanceof` type guard
- Truthiness narrowing
- Equality narrowing, for example: `===`

Error Handling

- In a `try...catch` statement, TypeScript infers errors as `unknown`
- Type for error in a `catch` block can only be declared as `any` or `unknown`

Flexible Types

`export` and `import`

`typeof` Type Operator

Intersection Types: &

Generics

- Generic functions
- Generic classes
- Generic types

Utility Types

- `Partial`, `Required`, `Readonly`, `Pick`, `Omit`, `Record`
- ...and more!

Debugging Type Errors

- **Tip:** 'Restart TS Server' command in Visual Studio Code
- Look up by error code: https://typescript.tv/errors/
- Fixes for common errors
  - Type assertion with `as` — lie to TypeScript!
  - Const assertion with `as const` — narrows the types

Tools & Resources

- TypeScript Cheat Sheets
- Pretty TypeScript Errors VSCode Extension
- Total TypeScript VSCode Extension
- Twoslash Query Comments VSCode extension